

NITHIN MATTAM

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SUMMARY

Passionate Engineer with 2 years of experience in Python, Java, SQL. Experienced in developing scalable applications, automating ETL data pipelines and, creating interactive dashboards. Passionate about building efficient systems that transform data into impactful insights.

EDUCATION

Master's in Data Science, University at Buffalo - The State University of New York **Aug'23 – May'25**

Bachelor's in Computer Science, Jawaharlal Nehru Technological University **Aug'19 – Jun'23**

SKILLS

Programming & Development: Python, Java, SQL, Data Structures, Algorithms, OOD, API, Unit Testing

Cloud: AWS (S3, Lambda, Glue, Redshift), GCP (BigQuery, Dataflow), Azure, Apache Spark, Databricks

Data Warehousing: ETL Pipelines, Apache Airflow, Snowflake, Data Modeling, DBT, Data Lake, Kafka

Data Analysis & Visualization: EDA, Data Preparation, Tableau, Power BI, Pandas, NumPy, Excel

Tools & Collaboration: Git, Docker, Kubernetes, Jira, Confluence, Agile/Scrum, CI/CD, Linux Scripting

EXPERIENCE

Research Analyst, Dept. of Biostatistics – University at Buffalo **Jan'25 – Current**

- Research focused on retaining infection control, COVID-19 like outbreaks using network-based strategies tracing secondary contact while reducing quarantine spikes and minimizing social disruption.
- Computed influence scores using PRINCE algorithm, flagged top 25% as “influential”, and targeted tracing top-k direct contacts and all direct + top-k secondary contacts instead of quarantining everyone.
- Simulated 150-day outbreaks on Haslemere and ER/BA networks, showing that prioritizing high-influence contacts reduced quarantines by 50% while still controlling infections at sustainable levels.
- Enhanced outbreak model stability by 30% while maintaining quarantine limit <20% of the population by extending PRINCE algorithm with adaptive top-k selection (10–50%) validated using perturbation tests.
- Improved early-case capture by 12% and reduced quarantines by 20% by introducing Bayesian-Network priors seeding PRINCE algorithm instead of a uniform-risk assumption at same workload.

Data Engineer, Saras Analytics **May'22 – Jul'23**

- Migrated 8+ legacy ETL jobs into Apache Airflow DAGs, improving job reliability by 20% and enabling data availability for client dashboards within hourly refresh windows.
- Designed and implemented a SQLAlchemy-based data quality rule engine, validating schema integrity and transformation logic, reducing downstream client-reported data issues by 35%.
- Developed relational data models for analytics pipelines, optimizing schema design for reporting tools and improving query efficiency by 25% across 50M+ records.
- Integrated ingestion workflows with Google Cloud Storage and BigQuery, achieving 99.9% pipeline uptime and enabling stakeholders to access updated datasets on demand.
- Cut incident resolution times by 40% by deploying River ML models (GaussianScorer, HalfSpaceTrees, OneClassSVM) to flag schema drift, spend spikes, and replication lag across marketing/order feeds.
- Collaborated with senior engineers and client teams in Agile sprints to refine ETL architecture, align deliverables with KPIs, and document workflows for seamless project handovers.

Projects

Distributed Data Processing with Apache Spark | Python, Jupyter Notebook, Pyspark, ML, Hadoop

- Built a distributed pipeline on Spark to process 1.2M Amazon book reviews (5.6 GB), applying structured transformations, imputing 100k+ missing values, and filtering IQR-based outliers to improve feature distributions and boost classification accuracy by 15%.
- Trained 3 ML models (logistic regression for spam review detection, linear regression for rating prediction, and k-means for review clustering) using Spark MLlib pipelines; achieving 91% accuracy and 18% RMSE reduction after cross-validation and hyperparameter tuning.

Multi-Platform Sentiment Analytics Pipeline | Python, AWS (S3, Athena, Lambda), Airflow, NLP

- Developed an end-to-end pipeline to track brand reputation and deliver real-time sentiment analysis by unifying data from 4+ social media API's - YouTube, Twitter, Reddit, Instagram, and News APIs.
- Orchestrated ETL with Airflow and Python to ingest 10k+ posts/day from APIs, applied schema validation and transformations, stored curated data in AWS S3 & Parquet, enriched with NLP sentiment scoring, and delivered dashboards with <15 min data freshness.